National University

of Computer and Emerging Sciences

Chiniot-Faisalabad Campus

Department of Artificial Intelligence and Data Science

Course Code	AI4001
Course Title	Fundamentals of Natural Language Processing
Assignment	1
Due Date	Monday, 02 March 2026, 08:00 PM

Instructions:

- Share .ipynb and .pdf file renamed as LT1_24F0001
- Submit task on Due Date. No Late Submission will be Tolerated.

Title

Customer Reviews Intelligence System using NLP and Machine Learning

Scenario

An e-commerce company receives a large number of customer reviews, chat messages, and support tickets daily. Recently, management has observed that customer feedback is becoming more challenging to analyze due to:

- Mixed English and Roman Urdu text
- Very short and informal reviews
- Messages where sentiment and intent do not align (e.g., positive sentiment but refund request)
- Important complaints being hidden among neutral feedback

The company wants an automated NLP-based system that can analyze this textual data and provide meaningful insights for decision-making.

You are required to design and implement an NLP solution that processes raw customer feedback and generates analytical outputs.

Tasks

1. Perform appropriate text preprocessing on the given customer reviews dataset, including cleaning, normalization, tokenization, and stop word handling. Show sample text before and after preprocessing.
2. Convert textual data into numerical representation using Bag of Words and TF-IDF, and compare their impact on model performance.
3. Implement sentiment analysis using:
    a. One rule-based approach
    b. One machine learning model (Naïve Bayes or Logistic Regression)
4. Perform intent classification to categorize customer messages into relevant classes such as:
    a. Complaint
    b. Refund Request
    c. Delivery Issue

d. General Query

5. Apply a topic modeling technique (e.g., NMF) to identify major themes discussed in customer feedback and interpret the results.
6. Evaluate model performance using Precision, Recall, F1 Score, and Confusion Matrix, and justify which metric is most suitable for this problem.

System Requirement (Interface)

1. In addition to the notebook, students must develop a simple interface using Gradio that allows a user to:
    a. Enter a new customer review
    b. View predicted sentiment
    c. View predicted intent
    d. Display identified topic or key keywords
2. The Gradio interface should demonstrate how the developed NLP system can be used interactively in a real-world setting.
3. Note: Only Gradio is allowed for the interface implementation.

Dataset (Choose Any One)

1. https://huggingface.co/datasets/mmehul/ecommerce-reviews-sentiment
2. https://www.kaggle.com/datasets/nicapotato/womens-ecommerce-clothing-reviews
3. https://zenodo.org/records/17612521
4. https://www.kaggle.com/datasets/snap/amazon-fine-food-reviews

Students may optionally add a small number of manually written noisy reviews (e.g., Roman Urdu or short complaints) to simulate real-world data.

Important Instructions

- Students are allowed to use any AI tools (e.g., ChatGPT, Copilot, etc.) while working on this assignment.
- Evaluation will be strictly viva-based, and students must clearly explain:
    - Preprocessing choices
    - Feature representation
    - Model selection
    - Evaluation metrics and results
- Use of AI tools without proper understanding will result in loss of marks.

Submission

Students must submit:

1. Jupyter Notebook containing complete implementation with pdf
2. Short technical report (2–3 pages) explaining methodology and results
3. Working Gradio interface demonstrating the system

Expected Learning Outcome

Students will be able to design an end-to-end NLP pipeline, apply vectorization and classification techniques, evaluate model performance, and demonstrate a functional NLP system using Gradio.

GOOD LUCK 😊